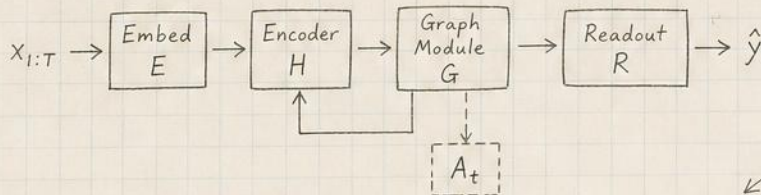
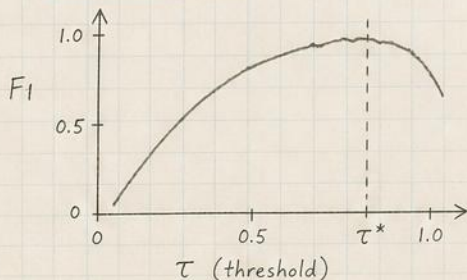


Project: Seq GraphGoal: learn structured representations for event sequences.Model sketch:Hypothesis:

If the graph module captures causal links, then performance improves near the decision threshold.

Threshold behavior (dev set):

$\tau^* \approx 0.72$
sweet spot.

Too low $\rightarrow$
FP $\uparrow$

Too high $\rightarrow$
FN $\uparrow$

Note to self:
revisit loss
weighting?
class imbalance
still an issue.

Loss (training): $L = \text{BCE}(\hat{y}, y) + \lambda \cdot \|A_t\|_F$
 $\lambda = 1e-3$ (current)

Quick calc (sanity):

$$0.72^2 + 2 \cdot 0.28 = 0.5184 + 0.56 = 1.0784$$

ignore,
wrong
term.

Next:

- sweep τ on test
- ablate G
- try temp. scaling

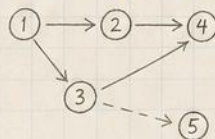
Results (dev set)

Model	τ	Prec.	Rec.	F1	AUROC
Base (no G)	0.50	0.62	0.48	0.54	0.71
SeqGraph ($\lambda=1e-3$)	0.72	0.71	0.68	0.69	0.82
SeqGraph ($\lambda=1e-4$)	0.72	0.69	0.65	0.67	0.80
SeqGraph ($\lambda=1e-2$)	0.72	0.63	0.60	0.61	0.77

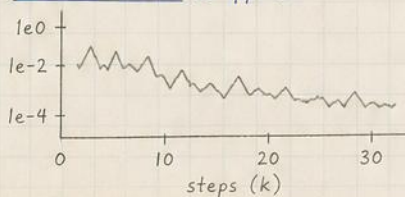
Best F1
at $\tau \approx 0.72$
(matches plot)

Error breakdown ($\tau = 0.72$):

$$\left. \begin{array}{l} TP = 342 \\ FP = 146 \\ FN = 160 \\ TN = 852 \end{array} \right\} \begin{array}{l} P = 0.71 \\ R = 0.68 \\ F1 = \underline{\underline{0.69}} \end{array}$$

 A_t example ($t=7$)

→ strong
--> weak

Gradient norm (snippet):

consider clipping
after 20k?
(instability spikes)

Ideas:

- edge dropout
- positional enc.?
- contrastive loss

